

HW05 Performance Test Report

Student ID 23127326 | Local run date 2026-08-30

1. Execution context

Field	Value
SUT	http://localhost:3000 commit 85af3ba875c88283615e22cb108f13e2fccaf0e9
Host	MacBook Pro 18,3 Apple M1 Pro 10 cores 32 GB RAM macOS 26.5.2
Tool	Apache JMeter 5.6.3 Java 21 headless non-GUI run

2. Workflow

Lockout probe -> valid login -> product search -> cart add -> cart quantity update -> cart read -> checkout -> post-checkout cart assertion. Each official VU uses a distinct synthetic account and a per-VU CSV file.

3. Scenario results

Scenario	Samples	HTTP err	Assertion gap	p95 overall / max label	TPS
Load	3,314	0 (0.00%)	359 (10.83%)	5 / 7 ms	9.2766
Stress	16,519	0 (0.00%)	1,789 (10.83%)	6 / 8 ms	34.5021
Spike	7,175	0 (0.00%)	753 (10.49%)	6 / 9 ms	17.1743
Endurance	24,608	0 (0.00%)	2,700 (10.97%)	5 / 7 ms	30.4317

Weighted mean: Load 2.435 ms; Stress 2.297 ms; Spike 2.437 ms; Endurance 1.888 ms. Full label-level p50/p90/p95/p99/max is in report/metrics-20260830/*.json.

4. Interpretation

All failed samples are the explicit POST_CHECKOUT_CART - expected empty assertion. The SUT leaves the cart non-empty after checkout. Therefore the 10.49-10.97% failure rates are business-gap evidence, not transport or HTTP failure. Response times are below the provisional 1,000 ms p95 threshold.

Resource conclusion is withheld. The official monitor used macOS-invalid ps field thcount and produced header-only files. The monitor tool is fixed to count threads with ps -M and validated with a real process sample, but that sample is not attributed to the four workload runs.

5. Reproduced SUT gaps

Gap	Evidence
Lockout	Four responses: 401, 401, 403, 403; DB attempts 4; locked window 180 s. See evidence/issues/lockout-probe-20260830.jsonl.
Checkout cleanup	Post-checkout assertion fails for every observed checkout in all four official workloads.
Pagination / duplicate line	Implementation candidates retained; independent response probe still required before external issue.

6. Reproduction and evidence

Raw JTL and JMeter HTML are under results/{load,stress,spike,endurance}. Plans are under test-plans. Synthetic data and provisioning scripts are under data/ and tools/. AI audit, critique and continuous-testing proposal are included. Video and GUI screenshots remain manual tasks for the student.